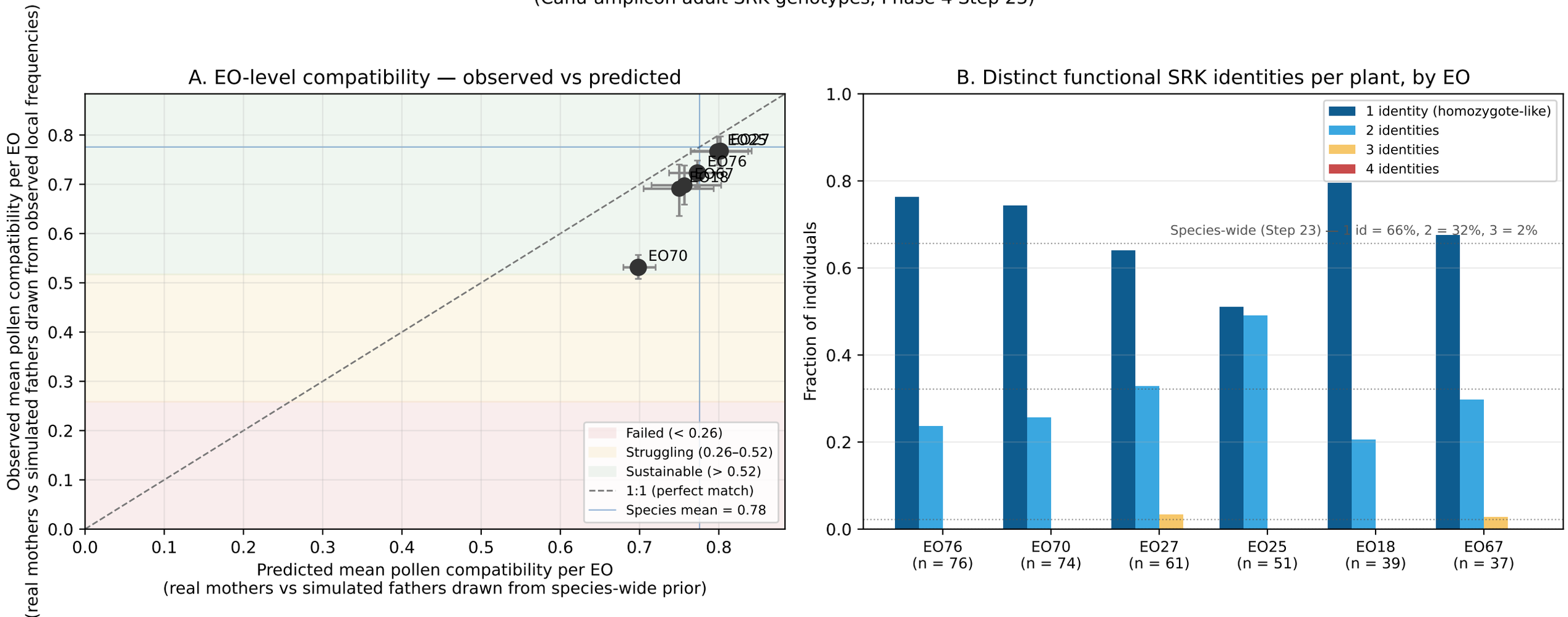
Step 30 Part C — First empirical validation of the sporophytic Class I / II + empirical-zygosity P_compat model at EO scale
(Canu-amplicon adult SRK genotypes, Phase 4 Step 23)
A. EO-level compatibility — observed vs predicted
Observed mean pollen compatibility per EO
(real mothers vs simulated fathers drawn from observed local frequencies)
Predicted mean pollen compatibility per EO
(real mothers vs simulated fathers drawn from species-wide prior)
EO27
EO76
EO67
EO18
EO70
Failed (< 0.26)
Struggling (0.26–0.52)
Sustainable (> 0.52)
1:1 (perfect match)
Species mean = 0.78
B. Distinct functional SRK identities per plant, by EO
Fraction of individuals
1 identity (homozygote-like)
2 identities
3 identities
4 identities
Species-wide (Step 23) — 1 id = 66%, 2 = 32%, 3 = 2%
EO76
(n = 76)
EO70
(n = 74)
EO27
(n = 61)
EO25
(n = 51)
EO18
(n = 39)
EO67
(n = 37)